

SARA AVILA

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EDUCATION

Creighton University

Aug 2023 – May 2027

B.S. in Computer Science, B.S. in Data Science, B.A. in Spanish and Hispanic Studies

GPA: 4.0/4.0

Related courses: Machine Learning, Introduction to Artificial Intelligence, Building Generative AI Applications, Data Structures, Algorithm Design and Analysis, Computer Organization, Programming Languages

RESEARCH EXPERIENCE

Advanced Deep Learning for Automated Cochlear Hair Cell Detection

2024 - 2026

Advisor: Dr. Steven Fernandes

Creighton University

- Cochlear hair cell quantification is traditionally performed through manual microscopy, limiting the speed and scalability of hearing loss research. This project develops deep learning methods for automated detection and classification of cochlear hair cells.
- Collected and annotated confocal microscopy images, trained YOLO-based object detection models, and evaluated detection performance against expert annotations to improve automated identification of inner and outer hair cells.

SCRIBE: Transformer-Based Evaluation of Medical School Personal Statements

2025 – 2026

Advisor: Dr. Steven Fernandes, Dr. Faith Kurtyka

- Medical school personal statements are often evaluated subjectively, creating inconsistent feedback for applicants. This project develops an offline NLP system that segments essays, classifies rubric-aligned content, and generates structured feedback.
- Organized and annotated training essays, evaluated transformer embedding models, and refined semantic segmentation and rubric-based classification to improve automated feedback aligned with medical school admissions criteria.

Enhancing Pharmacy Skills Through AI-Driven Patient Counseling Simulations

2024 – 2025

Advisor: Dr. Steven Fernandes, Dr. Jessica Cumber, Dr. Kevin Fuji

- Limited faculty availability restricts pharmacy students' opportunities to practice patient counseling with consistent feedback.
- Fine-tuned and evaluated LLM options to generate realistic, rubric-aligned patient responses, and designed the simulated patient's profile to support standardized counseling practice.

PUBLICATIONS

Krudwig, C., **Avila, S.**, Mondiri, A., Dhuler, A., Phillips, S., Viereck, A., Handel, K. V., and Fernandes, S. L. "GANs and YOLOv11 for Automated Cochlear Hair Cell Detection." *Thirteenth Conference on Frontiers of Intelligent Computing: Theory and Applications (FICTA)*, 2025. **Best Paper Award.**

INDUSTRY EXPERIENCE

Quality Assurance Engineer, Hudl

Lincoln, NE

Youth Club High School Coaches Department

May 2026 – Aug 2026

- Drove quality engineering across Hudl's American football platform through .NET modernization, repository-wide automated test coverage, code-level failure investigation, and cross-functional product decisions supporting releases used by hundreds of thousands of coaches and athletes

Software Engineer Intern, Aerostar International

Sioux Falls, SD

Stratospheric Balloon Team

May 2025 – June 2025

- Designed and built full-stack simulators for stratospheric balloon launches, developing front-end interfaces and backend services to model balloon performance and launch scenarios

Apple MacBook Technician, Apple
Computer Repair Team

Omaha, NE
Dec 2023 – Jan 2025

- Received Apple MacBook Technician Certification, diagnosing and repairing Apple hardware and software, including component replacements, troubleshooting, and system restorations.

PRESENTATIONS & TALKS

Advanced Deep Learning Integration for Detection of Missing Cochlear Hair Cells

Frontiers of Intelligent Computing: Theory and Applications (FICTA)	2025
National Conference on Undergraduate Research (NCUR)	2025
Women in Machine Learning (WiML) Workshop at NeurIPS	2025
Women in Machine Learning (WiML) Workshop at NeurIPS	2024

Evaluating Medical School Personal Statements with SCRIBE: A Fine-Tuned Transformer System

Women in Machine Learning (WiML) Workshop at ICCV	2025
Women in Machine Learning (WiML) Workshop at NeurIPS	2025
Nebraska Academy of Sciences	2025

Enhancing Pharmacy Student Skills Through AI-Driven Patient Counseling

Rush University AI Symposium on Healthcare Education	2025
Nebraska Academy of Sciences	2025

Interpretable Deepfake Detection

28th NSF EPSCoR National Conference	2024
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AWARDS

Barry Goldwater Scholar	2026
Best Paper Award, FICTA	2025
Best Undergraduate Humanities Poster Presentation, Creighton University	2025
Alpha Sigma Nu Jesuit Honor Society	2025
S.J. Bucko Scholarship	2023 – 2024
Next Generation Leaders Program Scholar	2023 – 2027
Dean's List, Creighton University	2023 – 2027

TEACHING EXPERIENCE

Teaching Assistant (CSC121)

Spring 25', Spring 26', Fall 26', Fall 27'

Instructors: David Reed, Catherine Baker, Sherri Weitz-Harms

- Teaching assistant for Computers and Scientific Thinking
- Hosted office hours, attended lab sessions, mentored course projects, and graded assignments

1:1 Tutor

Fall 24', Spring 25'

Office of Academic Success, Creighton University

- Hosted and taught 1:1 meetings across a wide range of CS classes, spanning both lower and higher level coursework

Leader of the Computer Repair Program

Fall 2024

iJay Practicum, Creighton University

- Taught and mentored new technicians on hardware repairs and diagnostics, helping them earn their Apple certification

LEADERSHIP & SERVICE

Peer2Peer Mentor , TRiO Student Support Services (SSS)	2026 – 2027
Volunteer , DREAM Scholars Service Program	2024 – 2027
Member , Next Generation Leaders Applicant Review Committee	2024 – 2027
Decurion , Creighton University Ratio Studiorum Program	2025 – 2026
Peer Mentor , Creighton First-Generation Student Support Team	2024 – 2025
Volunteer , NeurIPS	2024

SKILLS

Programming	C/C++, Python, Java, HTML/CSS, JavaScript/TypeScript, R, C#
Machine Learning	PyTorch, Hugging Face, Pandas, NumPy, Scikit-learn, TensorFlow, Transformers
Frameworks & Tools	FastAPI, Vue.js, GraphQL, Docker/Rancher, Git, Github/Gitlab
Languages	English (Native), Spanish (Native)
Art & Design	@YOURUSERNAME